



Comparison of Effectiveness of Different Composting Methods and Compost's Impact on Growth of *Zea mays* L. and *Phaseolus vulgaris* L.

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Received: 19th Mar, 2026

Accepted: 10th May, 2026

Published: 30th June, 2026

ABSTRACT

Background: Organic household solid waste composting is an environmentally sustainable approach for reducing waste while improving soil fertility and crop productivity. This study compared four composting methods—pit, pile, bin, and vermicomposting and evaluated the effects of the resulting composts on the growth of *Zea mays* L. and *Phaseolus vulgaris* L. in Kathmandu, Nepal.

Method: Household organic waste was segregated and composted using four methods over four months, with temperature monitored throughout composting process. After compost maturation, physical properties (temperature, moisture content, and light intensity) and chemical properties (pH, EC, OC, N, P, K and C/N ratio) were analyzed. Plant growth experiments were conducted using maize and bean under different compost treatments. Data normality for above-ground plant height was verified before applying one-way ANOVA, followed by Tukey's post hoc test for pairwise comparisons (10 plants per treatment).

Results: Significant differences ($p < 0.05$) were observed among compost treatments for both crops. Vermicompost produced significantly greater maize height than pit and pile composts, while bin compost significantly outperformed pit compost. For beans, bin compost produced significantly greater plant height than pit and vermicompost. The highest maize biomass (10.80 g) was obtained with bin compost, whereas vermicompost produced the highest bean biomass (10.68 g). Pile compost contained the highest nitrogen content (0.92%).

Conclusion: These findings demonstrate that composting methods significantly influence compost quality and crop performance. Bin compost was the most effective for maize growth, whereas vermicompost was superior for bean biomass production. The study highlights household organic waste composting as a sustainable waste management strategy that enhances nutrient recycling, supports environmentally friendly agriculture, and improves crop productivity.

Keywords: Composting; Compost quality; Nutrients; Organic solid waste; Waste management.

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INTRODUCTION

Composting of organic waste is an environmentally friendly approach that converts biodegradable waste into a valuable soil amendment while reducing volume of waste disposed of in landfills. Compost improves soil fertility by supplying essential nutrients, enhancing soil structure, increasing water-holding capacity, and promoting sustainable crop production [1, 2]. The effectiveness of compost depends largely on raw materials used, their nutrient composition, compost maturity, and composting method employed [3]. Soil structure also plays a vital role in nutrient availability, water movement, and root development, and incorporation of compost improves soil aeration, drainage, moisture retention, seedling establishment, and erosion control [1, 4].

Compost supplies essential macronutrients, including nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), sulfur (S), carbon (C), oxygen (O), and hydrogen (H), as well as micronutrients such as iron (Fe), boron (B), chlorine (Cl), zinc (Zn), copper

(Cu), molybdenum (Mo), and nickel (Ni), all of which are required for healthy plant growth [5]. N is essential for chlorophyll formation and vegetative growth. While compost releases nitrogen gradually through mineralization, excessive nitrogen application may cause rapid vegetative growth, weak root systems, and leaf burn, whereas nitrogen deficiency results in chlorosis, stunted growth, and reduced protein synthesis [6, 7]. Because nitrate is highly mobile in soil, use of stable, well-matured compost minimizes nutrient leaching and groundwater contamination while improving nutrient uptake by plants [5].

Phosphorus is required for protein synthesis, cell division, energy transfer, and root development. Adequate phosphorus promotes healthy plant growth, whereas deficiency causes stunted growth, delayed maturity, poor seed and fruit development, and dark blue-green leaves [6]. Potassium is equally important for chlorophyll production, disease resistance, and tolerance to drought and cold stress. Potassium deficiency results in leaf scorching, browning, and weak stems, while compost serves as an important source of potassium for healthy plant de-

velopment [8, 9]. The carbon-to-nitrogen (C/N) ratio is an important indicator of compost quality. Although the optimum C/N ratio for composting generally ranges from 25:1 to 30:1, successful composting has also been achieved at lower ratios of approximately 15:1 [10]. Several studies have demonstrated beneficial effects of compost and balanced nutrient application on plant growth. [11] investigated domestic organic waste composting in Madhyapur Thimi, Bhaktapur, and reported that approximately 80% of household waste was biodegradable. Their study found that pile composting recorded the highest composting temperature, while vermicompost contained the highest nitrogen concentration (1.17%). The C/N ratio ranged from 8.09 to 16.40, indicating well-decomposed compost suitable for agricultural use.

The importance of balanced nutrient supply has also been demonstrated in fertilizer studies. [12] reported that higher NPK application significantly increased plant height, with the greatest growth observed under the highest fertilizer treatment (F3). Similarly, [13] found that the tallest plants were produced with an NPK application of 150:75:50 kg ha⁻¹. [14] observed the largest stem diameter under treatment D (120:75:60 kg ha⁻¹ NPK), while plant height increased significantly with increasing NPK levels. [15] further reported that moderate NPK fertigation (34N-42P-29K) significantly enhanced leaf area ratio (LAR) and specific leaf area (SLA), whereas lower NPK levels improved nutrient use efficiency through a higher net assimilation rate (NAR). Likewise, [16] demonstrated that compost and vermicompost produced from food waste significantly improved plant height, leaf number, leaf area, shoot length, internode length, and branch production in *Capsicum annuum*.

These findings indicate that compost quality and nutrient availability play critical roles in improving plant growth and productivity. However, physicochemical characteristics of compost vary depending on composting method, which may influence its effectiveness as an organic fertilizer. Therefore, evaluating different composting methods and their effects on compost quality and crop performance is essential for promoting sustainable organic waste management and agricultural production.

METHODS

The composting experiment was conducted at the Environmental Research and Training Centre, Baneshwor, Kathmandu, Nepal. Organic household waste was collected from nearby residential areas. Four composting methods were evaluated: pit composting, pile composting, bin composting, and vermicomposting. Brown (carbon-rich) and green (nitrogen-rich) wastes were mixed in equal proportions (50:50) for pit and pile composting. Bin composting used 42% brown waste, 42% green waste, and 16% soil, while vermicomposting consisted of 38% brown waste, 38% green waste, and 24% soil. Soil was added to improve moisture retention, aeration, microbial activity, and, in vermicomposting, to provide a suitable habitat for earthworms.

Pit Composting

A pit (1 ft × 1 ft × 1 ft) was prepared and filled with 2.5 kg each of brown and green waste. Moisture was maintained by periodic watering (Figure A2).

Pile Composting

A compost pile was constructed over a base of small twigs to improve aeration. Brown and green wastes (2.5 kg each) were layered, and the pile was turned regularly to enhance aerobic decomposition (Figure A3).

Bin Composting

A 100-L plastic bin was lined with newspaper and a thin layer of soil (2.37 kg). Brown and green wastes (6.12 kg each) were added, and moisture was maintained throughout the composting period (Figure A4).

Vermicomposting

A plastic basin was filled with 2.5 kg each of brown and green waste, 1.624 kg soil, and 1 kg of *Eisenia fetida*. The unit was covered with moist jute cloth to maintain suitable moisture and temperature for earthworm activity (Figure A5).

The effects of compost on plant growth were evaluated using maize (*Zea mays* L.) and bean (*Phaseolus vulgaris* L.). Twenty seeds of each crop were sown in polybags containing each compost treatment. Plants were maintained under natural sunlight with regular watering, and plant height was measured every two days (Figure A6).

Compost samples were analyzed for pH, electrical conductivity (EC), total nitrogen (TN), total phosphorus (TP), total potassium (TK), and organic carbon (OC). pH and EC were measured using digital meters. Total nitrogen was determined by the Kjeldahl method, total phosphorus by nitric-perchloric acid digestion followed by the molybdovanadate spectrophotometric method, total potassium using a flame photometer, and organic carbon by the Walkley-Black wet oxidation method.

Data were analyzed using Microsoft Excel and IBM SPSS Statistics version 25. Descriptive statistics were calculated for all variables. Normality was assessed using the Shapiro-Wilk test. Differences in plant height among composting methods were evaluated using one-way ANOVA followed by Tukey's HSD post hoc test [17, 18] when significant differences were detected ($p < 0.05$). Statistical significance was set at $\alpha = 0.05$.

RESULTS

Physicochemical Properties of Compost

The temperature of compost was monitored throughout the composting period for all four composting methods. Linear regression analysis showed a strong relationship between composting time and temperature in each method, with R^2 values greater than 0.80. This indicates that more than 80% of the variation in compost temperature was explained by the progression of composting time. The high R^2 values demonstrate a consistent and

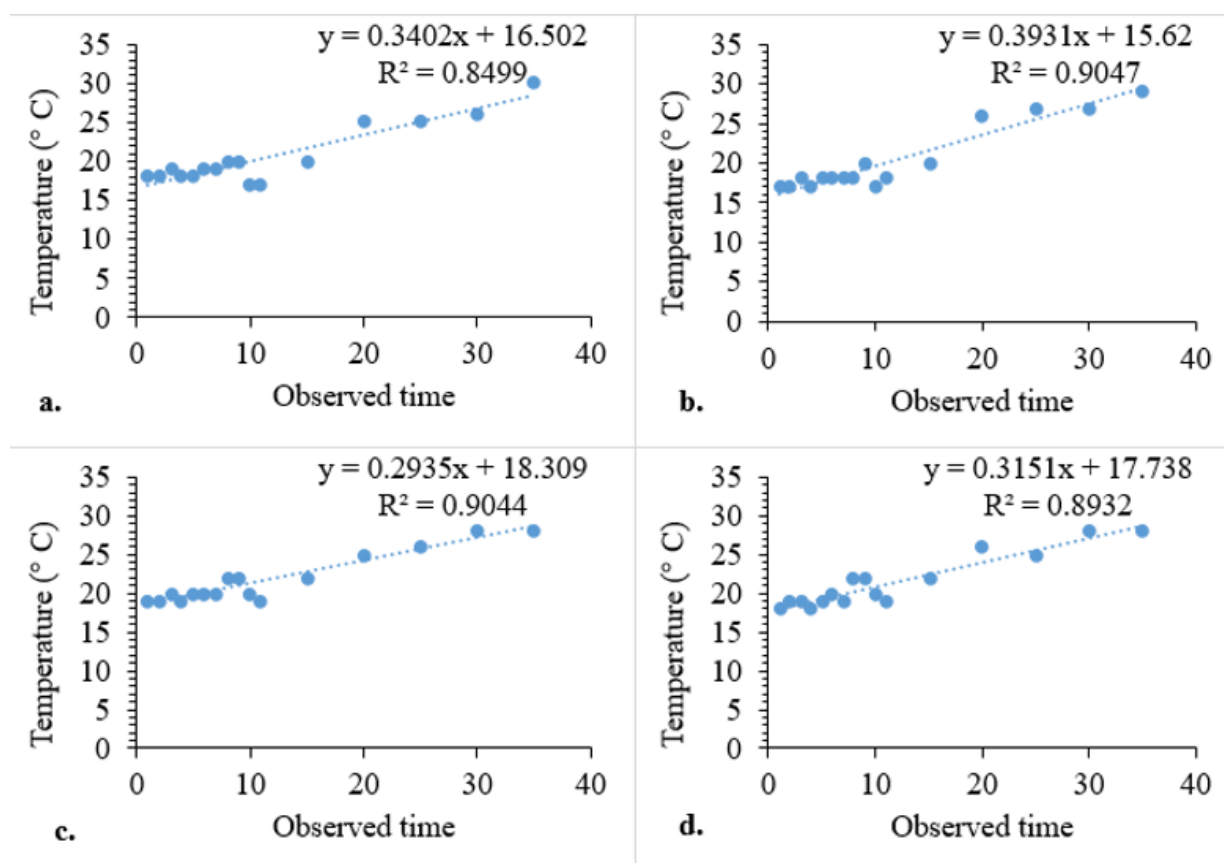


Figure 1. Change in temperature of compost a. pit b. pile c. bin d. vermi over time (linear regression)

Table 1. Physicochemical properties of compost

Compost type	L.I	Temp. (°C)	Mo. (%)	pH	EC	OC (%)	TN (%)	TP (%)	TK (%)
Pit	2750	30	25	8.0	0.24	15.04	0.15	0.27	0.13
Pile	42500	29	2.5	9.1	0.69	21.89	0.92	0.30	0.13
Bin	250	28	45	9.0	1.18	12.57	0.69	0.18	0.14
Vermi	250	28	50	8.2	0.75	11.24	0.77	0.19	0.13

predictable temperature pattern during the composting process, suggesting efficient decomposition and the effectiveness of all four composting methods Table 1 and Figure 1.

The physicochemical properties varied among the four composting methods Table 1 and Figure 1. The highest peak temperature was recorded in pit compost, followed by pile compost, whereas bin and vermicompost exhibited the same peak temperature. Moisture content was highest in vermicompost and lowest in pile compost, likely due to greater aeration under open conditions. The highest pH was observed in pile compost (9.1), indicating greater alkalinity, while pit compost had the lowest pH (8.0). Organic carbon content was highest in pile compost (21.89%) and lowest in vermicompost. Total nitrogen was also highest in pile compost (0.92%), followed by vermicompost (0.77%), bin compost (0.69%), and pit compost (0.15%) (Figure 1). Similarly, total phosphorus was highest in pile compost (0.30%) and lowest in bin compost (0.18%). In contrast, bin compost contained the highest total potassium content (0.14%), whereas vermicompost, pit compost, and pile compost each contained

0.13% potassium.

Comparison of Aboveground Height of Plant

A one-way analysis of variance (ANOVA) was conducted to evaluate the effect of different compost types on the aboveground height of two plant species, maize and bean. For maize, the ANOVA produced an F -value of 5.075 with a p -value of 0.003, indicating a statistically significant difference in plant height among the compost treatments. Similarly, for bean, the analysis yielded an F -value of 8.680 with a p -value < 0.001 , demonstrating a highly significant effect of compost type on plant height. These findings indicate that compost type significantly influenced the aboveground height of both maize and bean plants.

After the ANOVA showed a significant difference in plant height, Tukey's post hoc test was used to analyze pairwise comparisons between group means to identify which specific groups had significant differences ($p < 0.05$). There was a significant difference between Pit and Bin, Pit and Vermi, and Pile and Vermi compost.

Plant Biomass



Figure 2. Live Biomass of maize and bean in different composts

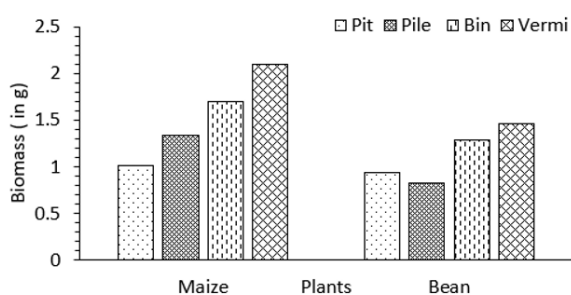


Figure 3. Dry biomass of maize and bean in different compost.

Plant biomass was determined by measuring both fresh and dry biomass. Fresh biomass was recorded by weighing the entire plant immediately after harvesting, before drying. The harvested plants were then oven-dried at 60 °C for 48 hours to remove moisture. After drying, the plants were weighed again to obtain their dry biomass. The dry biomass provided a more accurate measure of plant growth, as it represented the plant's actual biomass without the influence of water content Figures 2 and 3.

DISCUSSION

The physicochemical properties of compost are important indicators of compost quality and nutrient availability. In present study, pile compost contained the highest nitrogen concentration, followed by vermicompost. Although [11] reported higher nitrogen content in vermicompost, differences in feedstock composition may explain the variation. The relatively lower nitrogen content in vermicompost may also result from nitrogen assimilation by earthworms during growth and reproduction.

Vermicompost had the lowest C:N ratio (14.6), indicating greater compost maturity and nutrient availability, whereas pit compost had the highest C:N ratio (100.2), suggesting slower decomposition. Similar findings have been reported by [19], who attributed the lower C:N ratio in vermicompost to enhanced microbial activity and earthworm-mediated decomposition.

EC was highest in bin compost, likely due to improved aeration and greater mineralization of organic matter, while pit compost showed the lowest EC because of re-

duced aeration and nutrient leaching [20]. Vermicompost exhibited the highest moisture content, reflecting its finer structure and greater water-holding capacity [21]. Pile compost had the highest pH due to efficient aerobic decomposition, whereas pit compost remained more acidic because of anaerobic decomposition and organic acid accumulation [22, 23].

Plant growth and biomass are strongly influenced by nutrient availability. Previous studies have shown that balanced nutrient levels promote better growth than either excessive or deficient nutrient supply [15, 24]. In this study, vermicompost produced the highest biomass in both maize (*Zea mays* L.) and bean (*Phaseolus vulgaris* L.). This superior performance is likely due to its balanced nutrient content, enhanced microbial activity, improved moisture retention, and presence of plant growth-promoting substances. These findings are consistent with [25], who reported significant improvements in plant biomass following vermicompost application.

CONCLUSION

The study concludes that compost type significantly influences plant growth and biomass, with its effectiveness depending on physicochemical properties such as nutrient content, C:N ratio, moisture, pH, EC, and the specific nutrient requirements of the crop. Vermicompost produced the highest growth and biomass in maize, while bin compost resulted in the greatest plant height in bean. Therefore, compost should be selected according to nutrient needs of target crop. For improved crop performance, vermicompost is recommended for maize cultivation, and combining different compost types may further enhance plant growth. Future studies should evaluate long-term effects of compost application on soil fertility and crop productivity.

Acknowledgements: We are grateful to the Environmental Research and Training Center (ERTC) for providing laboratory facilities and financial support during the fieldwork. We also sincerely thank Nepal Environmental and Scientific Services (NESS) for conducting the laboratory analyses of nitrogen, phosphorus, potassium, and organic carbon.

Conflict of Interest: None

Funding: None

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Citation: Luitel P, Deo R, Malla-Pradhan R, Karn L L. Comparison of Effectiveness of Different Composting Methods and Compost's Impact on Growth of *Zea mays* L. and *Phaseolus vulgaris* L.. *IJSIRT*. 2026;4(1):12–17. DOI: [10.3126/ijsirt.v4i1.XXXXX](https://doi.org/10.3126/ijsirt.v4i1.XXXXX)

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